Midterm Practice

- **Use the questions in this question bank to practice for the midterm test.**
- **Syllabus:** everything covered up to and including the week before the midterm test.
  Reminders:
- The midterm counts for **20%** of your final grade.
- Both the midterm and the final test are **closed book.**
  - However, you are allowed to bring 1 A4 size sheet.
  - You are also allowed to bring your calculator.

1. Which of the following is the correct way to create a NumPy array from a Python list?

- a) np.create_array([1, 2, 3])
- b) np.array([1, 2, 3])
- c) np.arr([1, 2, 3])
- d) np.make_array([1, 2, 3])

Answer: b

2. How can you create a 2x3 matrix using Python lists with NumPy?

- a) np.array([1, 2, 3, 4, 5, 6], shape=(2,3))
- b) np.array([[1, 2, 3], [4, 5, 6]])
- c) np.matrix([1, 2, 3], [4, 5, 6])
- d) np.array(2, 3, [1, 2, 3, 4, 5, 6])

Answer: b

3. What does np.linspace(0, 1, 5) return?

- a) An array with 4 evenly spaced elements between 0 and 1
- b) An array with 5 evenly spaced elements between 0 and 1
- c) An array with 6 evenly spaced elements between 0 and 1
- d) A scalar value

Answer: b

4. How do you create a 3x3 matrix of zeros in NumPy?

- a) np.zeros(3, 3)
- b) np.zeros((3, 3))
- c) np.zeros[3, 3]
- d) np.create_zeros((3, 3))

Answer: b

5. Which function is used to reshape a NumPy array?

- a) np.shape()
- b) np.resize()
- c) np.reshape()
- d) np.array_shape()

Answer: c

6. How can you find the maximum value in a NumPy array?

- a) array.maximum()
- b) np.maximum(array)
- c) array.max()
- d) max(array)

Answer: c

7. What does broadcasting refer to in NumPy?

- a) Copying elements from one array to another
- b) Extending smaller arrays during arithmetic operations
- c) Broadcasting values to external files
- d) Sharing data between different arrays

Answer: b

8. How can you slice the first two rows of a 2D NumPy array?

- a) array[1:2]
- b) array[0:2, :]
- c) array[:, 1:2]
- d) array[:2, 1]

Answer: b

9. Which of these selects all elements in an array greater than 5?

- a) array > 5
- b) array[array > 5]
- c) array.select(>5)
- d) select(array > 5)

Answer: b

10. What is the result of np.exp([0, 1, 2])?

- a) [0, 1, 4]
- b) [1, 2.71, 7.39]
- c) [1, e, e^2]
- d) Exponentiation is not possible

Answer: c

11. How do you compute the sine of an array in NumPy?

- a) np.sine(array)
- b) np.sin(array)
- c) sin(array)
- d) array.sin()

Answer: b

12. Which method computes the natural logarithm of each element of a NumPy array?

- a) np.log()
- b) np.nlog()
- c) np.ln()
- d) log(array)

Answer: a

13. What is the purpose of np.linalg.norm()?

- a) To compute the mean of an array
- b) To find the length of an array
- c) To compute the norm of a vector
- d) To determine the size of a matrix

Answer: c

14. How do you transpose a matrix in NumPy?

- a) np.transpose()
- b) matrix.T
- c) matrix.transpose()
- d) All of the above

Answer: d

15. What is the correct syntax to compute the inverse of a matrix?

- a) np.inverse(matrix)
- b) np.linalg.inverse(matrix)
- c) np.linalg.inv(matrix)
- d) matrix.inverse()

Answer: c

16. Which function is used to create a matrix of all ones?

- a) np.ones()
- b) np.create_ones()
- c) np.matrix_ones()
- d) ones.matrix()

Answer: a

17. How would you sum all the elements in a 2D array?

- a) array.sum(axis=None)
- b) np.sum(array)
- c) array.total()
- d) sum(array)

Answer: b

18. How can you change the shape of an array without modifying its data?

- a) np.resize()
- b) np.reshape()
- c) array.change_shape()
- d) array.reshape_copy()

Answer: b

19. Which operation does broadcasting NOT support?

- a) Adding a scalar to a matrix
- b) Multiplying two matrices of different shapes
- c) Element-wise division
- d) None of the above

Answer: b

20. How can you extract the minimum value index from an array?

- a) np.argmin(array)
- b) array.index_min()
- c) np.index_min(array)
- d) array.arg_min()

Answer: a

21. Which NumPy function returns an array of square roots?

- a) np.sqrt()
- b) np.sqroot()
- c) array.sqrt()
- d) np.square_root()

Answer: a

22. What does array[1:3, 2] return?

- a) The 1st and 2nd rows from the 2nd column
- b) The 2nd and 3rd rows from the 3rd column
- c) The 2nd and 3rd rows from the 2nd column
- d) None of the above

Answer: c

23. How do you initialize a 4x4 matrix of zeros?

- a) np.zeros(4, 4)
- b) np.zeros((4, 4))
- c) np.zeros[4x4]
- d) np.create.zeros((4,4))

Answer: b

24. Which method is used to reshape a (4, 4) array to a (2, 8) array?

- a) np.array_reshape(2,8)
- b) np.reshape(array, (2, 8))
- c) array.reshape(2, 8)
- d) array.reform(2, 8)

Answer: b

25. What is the L2 norm of the vector v = [3, 4]?

A. 5
B. 6
C. 7
D. 8

Answer: A

26. If u = [2, -3] and v = [4, 5], what is the dot product u · v?

A. 8
B. -6
C. -7
D. 7

Answer: C

27. If a vector is orthogonal to another, their dot product is:

A. Positive
B. Negative
C. Zero
D. Undefined
Answer: C

28. Which of the following describes the inverse of a matrix?

A. A matrix that when multiplied by the original gives the identity
B. A matrix that cancels the original to give the zero matrix
C. A matrix that equals the original
D. A matrix that equals the negative of the original
Answer: A

29. The transpose of a matrix A is obtained by:

A. Replacing rows with columns
B. Replacing columns with rows
C. Both A and B
D. Neither A nor B
Answer: C

30. If two matrices have the same dimensions, they can be:

A. Added
B. Multiplied
C. Divided
D. None of the above
Answer: A

31. What does the operation 'transpose' do to the matrix [[1, 2], [3, 4]]?

A. [[1, 3], [2, 4]]
B. [[3, 1], [4, 2]]
C. [[2, 1], [4, 3]]
D. None
Answer: A

32. The matrix [[1, 0], [0, 1]] is known as:

A. Diagonal Matrix
B. Zero Matrix
C. Identity Matrix
D. Symmetric Matrix
Answer: C

33. Given two matrices P and Q, the inverse of their product PQ is equal to:

A. $P^{-1} Q^{-1}$
B. $Q^{-1}P^{-1}$
C. The identity matrix
D. None of the above
Answer: B

34. How do you calculate the mean of the numbers 2, 4, 6, 8, and 10?

a) (2+4+6+8+10)/4
b) (2+4+6+8+10)/5
c) (2*4*6*8*10)
d) 2+4+6+8+10
Answer: b

35. What is the median of the data set: 3, 1, 4, 2, 5?

a) 2
b) 4
c) 3
d) 5
Answer: c

36. Which value appears most frequently in the data set: 7, 8, 8, 9, 10?

a) 7
b) 8
c) 9
d) 10
Answer: b

37. What is the first quartile (Q1) of the data set: 1, 2, 3, 4, 5, 6, 7, 8?

a) 2
b) 4
c) 1
d) 3
Answer: d

38. What does the third quartile (Q3) represent?

a) The maximum value in the data set
b) The median of the entire data set
c) The median of the upper half of the data
d) The average of all data
Answer: c

39. Which formula represents variance?

a) The square root of the standard deviation
b) The average of squared deviations from the mean
c) The difference between max and min
d) The sum of all data points
Answer: b

40. How is standard deviation related to variance?

a) It is the square of the variance
b) It is the square root of the variance
c) It is unrelated to variance
d) It is twice the variance
Answer: b

41. Which of these represents a measure of spread?

a) Mean
b) Mode
c) Standard Deviation
d) Median
Answer: c

42. What does a correlation coefficient of -1 indicate?

a) Strong positive relationship
b) Strong negative relationship
c) No relationship
d) Weak positive relationship
Answer: b

43. What is the formula for calculating the mean?

a) Total number of observations / Sum of observations
b) Sum of observations / Total number of observations
c) Sum of squares of deviations / Number of observations
d) Sum of maximum and minimum values
Answer: b

44. In a perfectly symmetrical distribution, how do mean, median, and mode usually compare?

a) Mean > Median > Mode
b) Mean = Median = Mode
c) Mean < Median < Mode
d) No particular relation
Answer: b

45. Which measure is most affected by outliers?

a) Mean
b) Median
c) Mode
d) First quartile
Answer: a

46. Which statement about variance is true?

a) Variance can never be negative
b) Variance can be negative
c) Variance is a measure of central tendency
d) Variance equals mean
Answer: a

47. How is covariance different from correlation?

a) Covariance is always between -1 and 1, correlation is not
b) Covariance is unbounded, correlation is between -1 and 1
c) Covariance measures non-linear relationships, correlation does not
d) There is no difference
Answer: b

48. What does a high positive covariance indicate?

a) The two variables move in opposite directions
b) The two variables do not move together
c) The two variables move in the same direction
d) The variables are negatively correlated
Answer: c

49. If the variance of a data set is 0, what does that imply?

a) The data set is highly spread out
b) There is no mean
c) All values in the data set are identical
d) There is no correlation
Answer: c

50. Which formula would you use to calculate the median?

a) Arrange data in ascending order and find the middle value
b) Divide the sum of data by number of observations
c) Count the most frequent value
d) Calculate the mean of all data points
Answer: a

51. What does the standard deviation measure?

a) The mean value of the data set
b) The spread or dispersion of the data set
c) The highest value minus the lowest
d) The skewness of the distribution
Answer: b

52. How can you calculate the first quartile (Q1)?

a) Find the middle value of the upper half of the data set
b) Find the middle value of the lower half of the data set
c) Subtract mean from median
d) Divide the range by 4
Answer: b

53. If the correlation coefficient is close to 0, what does it mean?

a) Strong relationship
b) Weak relationship or no relationship
c) Perfect negative relationship
d) Perfect positive relationship
Answer: b

54. How do you find the range of a data set?

a) Mean value - Median value
b) Maximum value - Minimum value
c) First quartile - Third quartile
d) Mode of the data
Answer: b

55. What does the median represent?

a) The middle value when data is sorted
b) The average of all numbers
c) The most frequent number
d) The maximum value
Answer: a

56. In statistics, what does the mode represent?

a) The highest value in the data set
b) The most frequently occurring value in the data set
c) The middle value
d) The average
Answer: b